

Mechanisms & Vibrations

- Transient Vibrations

Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

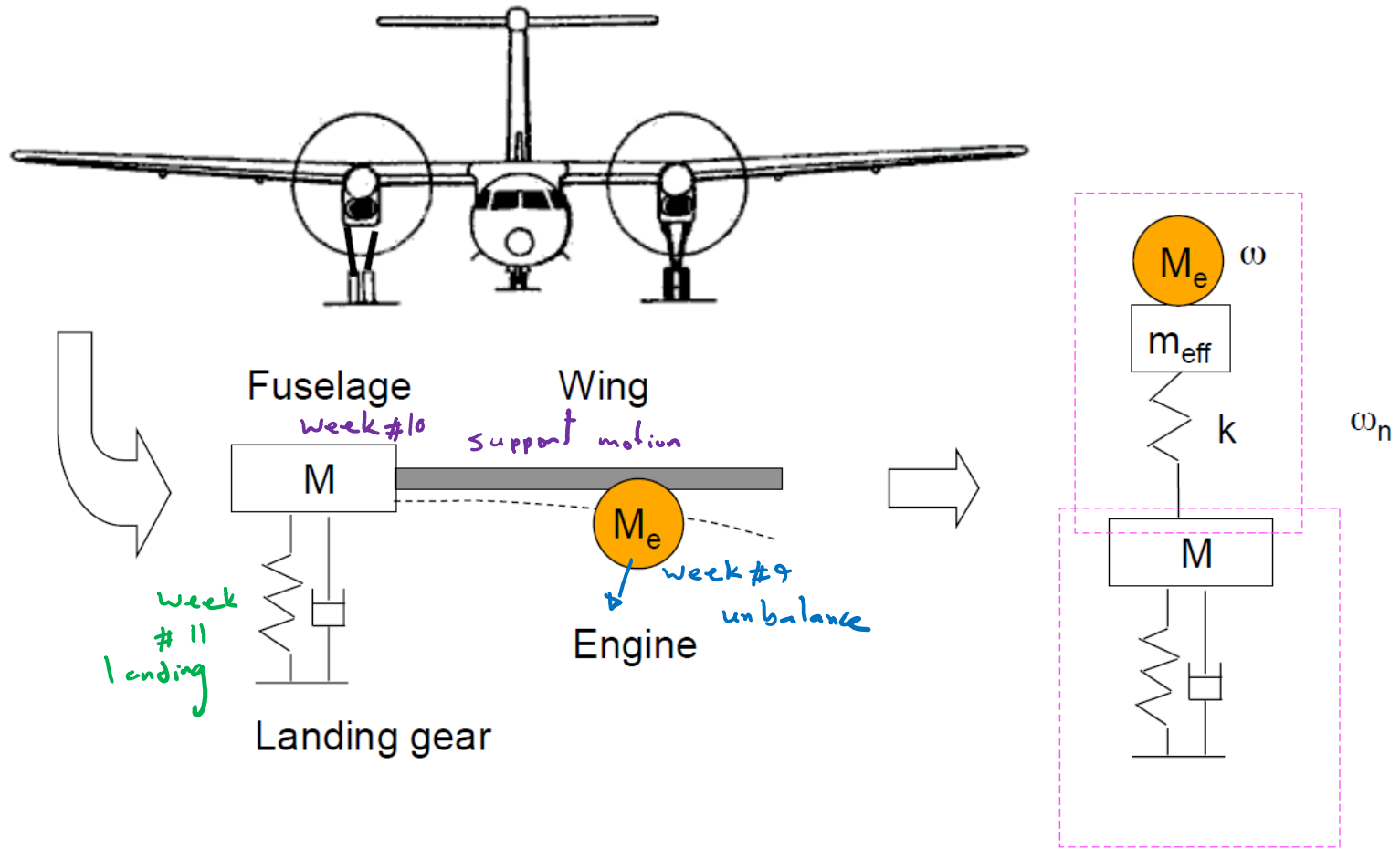
Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

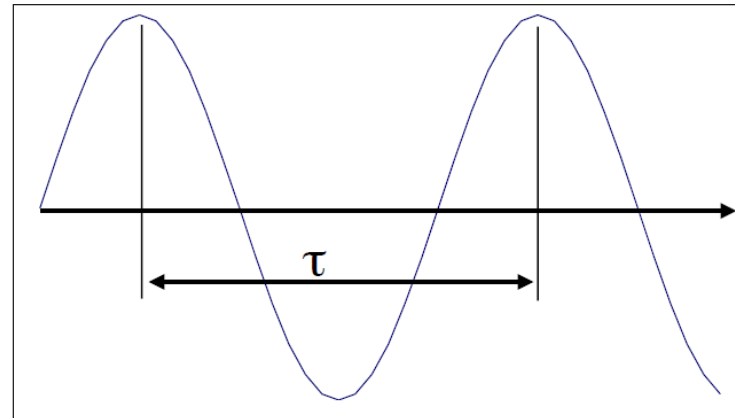
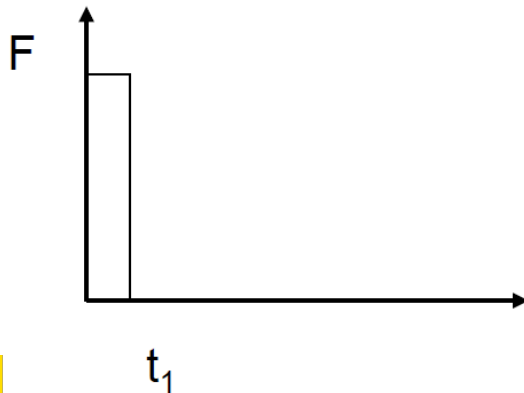
Simplified aircraft dynamics model



Introduction

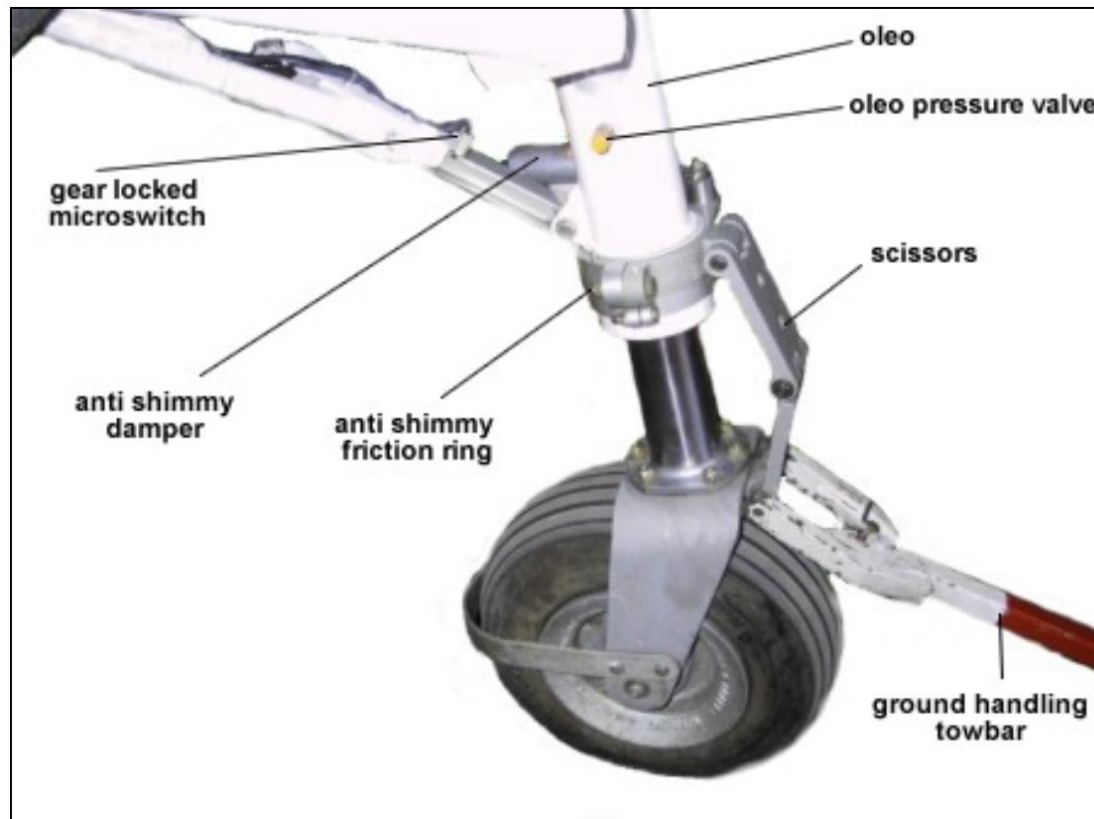
What is a shock?

- A shock is an impact.
- The duration of a shock is smaller than the natural period of the vibration system: $t_1 < \tau$ *excitation time less than period*
- A shock is also called pulse excitation.
- Aircrafts are subject to shock when landing.



Landing gears

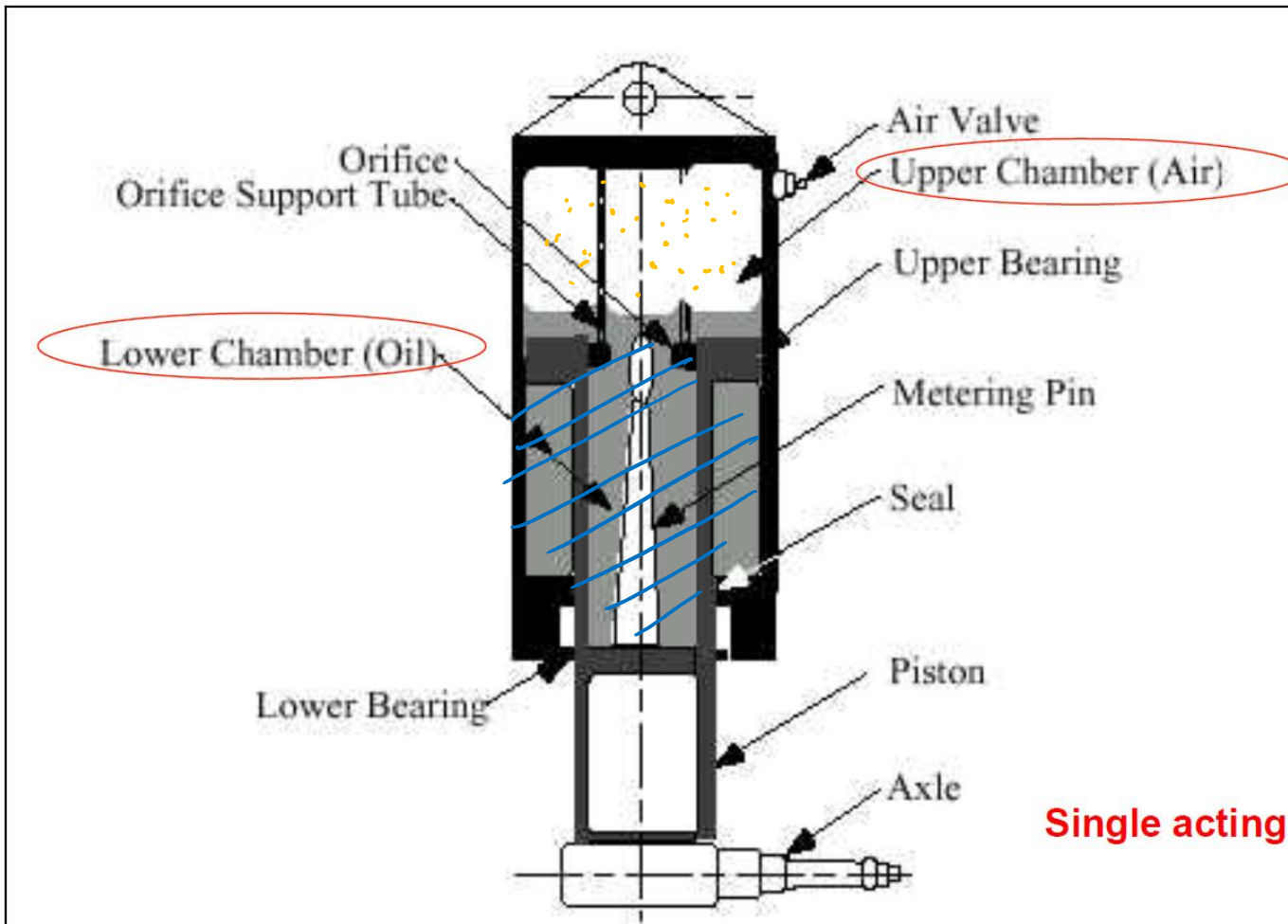
Landing gears are subject to impact when touchdown.



Shock absorber

- The basic function of the shock absorber is to absorb and dissipate the kinetic energy to the extent that accelerations imposed on the aircraft frame are reduced to a tolerable level ($0.4g = 12.9 \text{ ft/s}^2$, 10 ft/s, Boeing 747-400: 870k lb).
- Types: 1) solid spring made of steel and rubber;
2) fluid spring made of compressed gas or oil;
3) combined, called oleo shock absorber.
- Oleo uses compressed gas and oil to absorb and dissipate the impact energy.
- When landing, the strut and the piston is pushed together.
- This forces a chamber of oil (hydraulic fluid) against a chamber of gas (nitrogen), so both oil and gas are compressed to absorb the impact energy.
- After impact, the compressed gas will push the oil back to its chamber to dissipate the absorbed energy.

Oleo shock absorber



Impact and free vibration

- An impact is a force applied to a vibration system within a very short period of time:

$$I = F\Delta t \quad \text{or} \quad I = \int_0^{t_1} F(t) dt$$

- The impact is converted to linear momentum to excite vibration

$$I = mv(0) + mv(f) \rightarrow v(0) = I/m, \quad v(f) = 0$$

- Recall free vibration is

$$x = v(0)/\omega_n \sin \omega_n t + x(0) \cos \omega_n t$$

where $\omega_n = (k/m)^{1/2}$, $\tau = 1/f_n = 2\pi(m/k)^{1/2}$

- Under an impulse, free vibration is

$$x = v/\omega_n \sin \omega_n t$$

- If the impulse is considered as a (unit) pulse excitation, then

$$v(0) = 1/m \quad (\text{m/s or inch/s})$$

$$\rightarrow x = 1/m\omega_n \sin \omega_n t \rightarrow$$

$$\frac{v(0)}{\omega_n} \sin \omega_n t = \frac{\frac{I}{m}}{\omega_n} \sin \omega_n t \xrightarrow{I=1} \frac{1}{m\omega_n} \sin \omega_n t$$

week #18

$$x(t) = \frac{\dot{x}(0)}{\omega_n} \sin \omega_n t$$

$$= \frac{v(0)}{\omega_n} \sin \omega_n t$$

Arbitrary excitation & convolution

- An excitation with an arbitrary shape can be considered as accumulation of a series of impulse responses.
- The response to the unit pulse is defined

$$h(t) = 1/m\omega_n \sin\omega_n t$$

- The response to an arbitrary impulse is

$$x(t) = I(t) h(t - \xi)$$

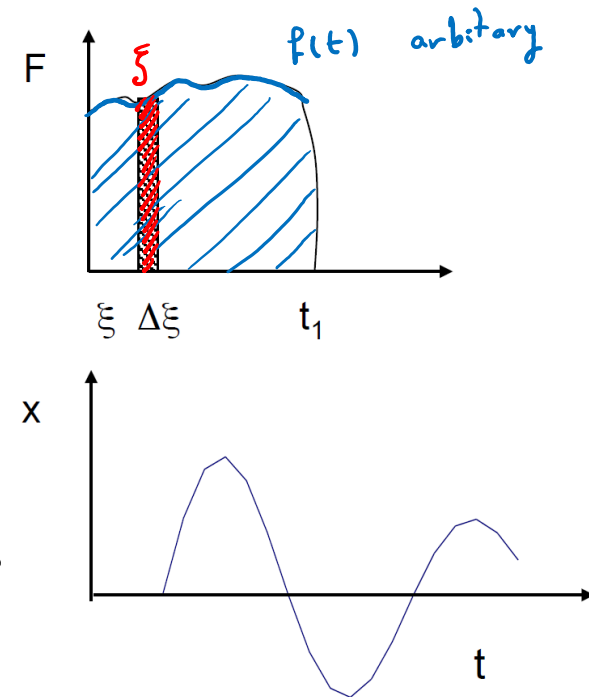
- Where an impulse is defined as

$$I(t) = F(\xi) \Delta\xi$$

- The response at time t to an arbitrary excitation is

$$x(t) = \int_0^t F(\xi) h(t - \xi) d\xi$$

- Convolution or superposition integral.



Example 11.1 step excitation

Determine the response of a spring-mass system to a step excitation

The impulse response is

$$\rightarrow h(t) = (1/m\omega_n) \sin \omega_n t$$

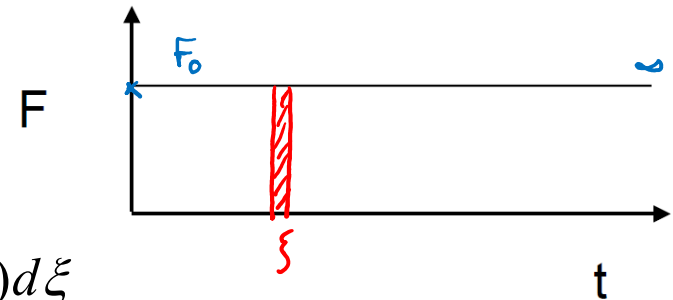
The response to a step excitation is

$$x(t) = \int_0^t F(\xi) h(t - \xi) d\xi = (F_o / m\omega_n) \int_0^t \sin \omega_n (t - \xi) d\xi$$

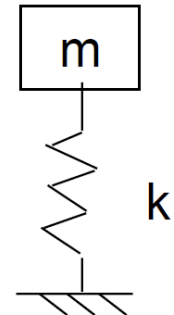
$$= (F_o / m\omega_n) \int_t^0 \sin \omega_n z (-dz) = (F_o / k)(1 - \cos \omega_n t)$$

Note the satisfied initial conditions are

$$x(0) = v(0) = 0$$



→ response of
a spring-mass
system under a
force $f(t) = F_o$



$$\xi \quad 0 \rightarrow t$$

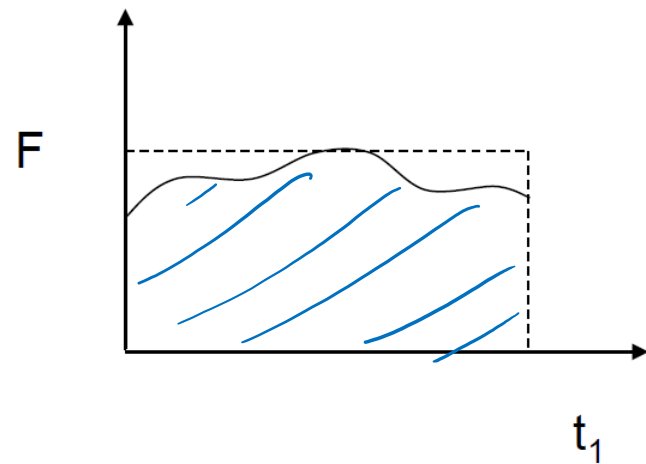
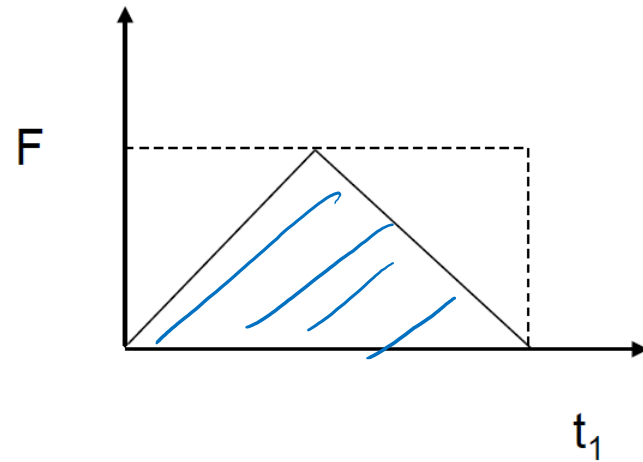
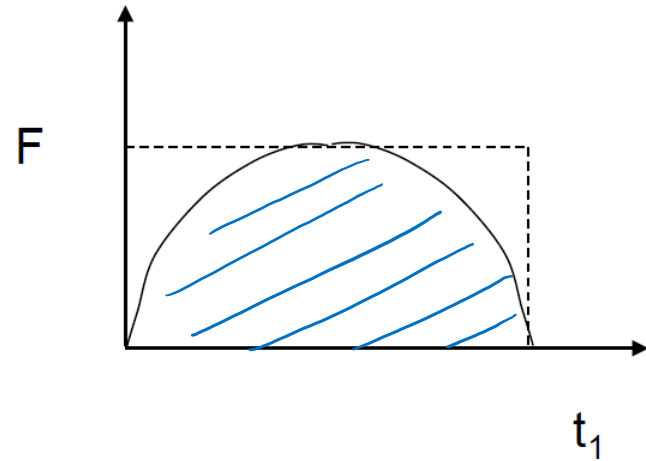
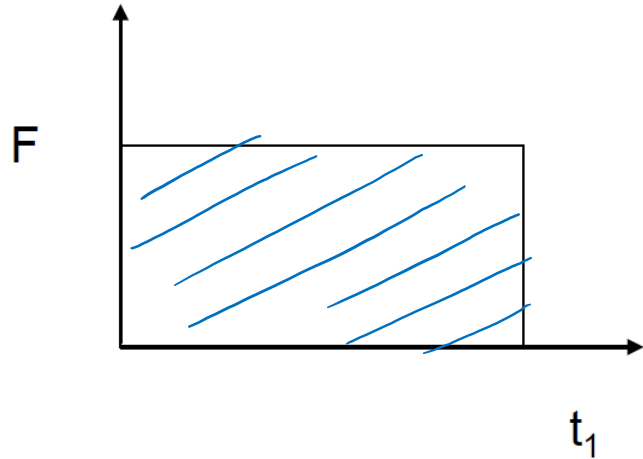
$$z \leftarrow t - \xi \quad t \rightarrow 0$$

$$t - \xi = z$$

$$-d\xi = dz$$

$$d\xi = -dz$$

Classical shock pulse shapes



Example 11.2 rectangular excitation

A rectangular excitation can be considered as the sum of two step functions.

1st response to a positive step

$$x_1 = (F_o/k)(1 - \cos \omega_n t) \quad (t < t_1)$$

2nd response to a negative step

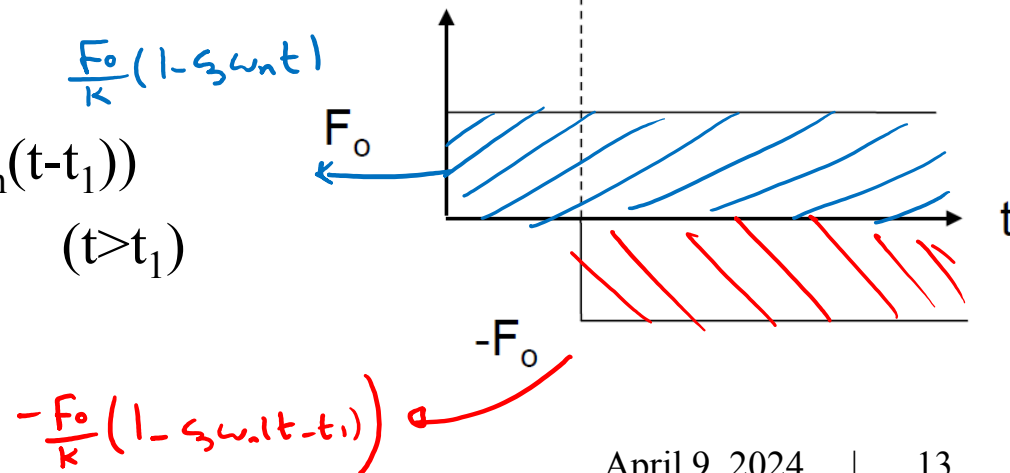
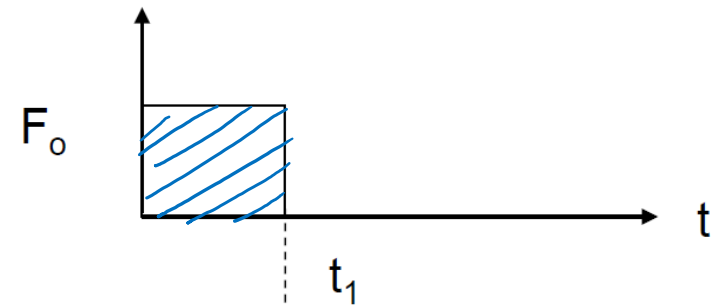
$$x_2 = (-F_o/k)(1 - \cos \omega_n (t - t_1)) \quad (t > t_1)$$

The total response is

$$x = x_1 + x_2$$

$$= (F_o/k)(1 - \cos \omega_n t) - (F_o/k)(1 - \cos \omega_n (t - t_1))$$

$$= (F_o/k)(-\cos \omega_n t + \cos \omega_n (t - t_1)) \quad (t > t_1)$$



Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

Shock response spectrum

- Shocks can be further divided into
 - Short duration pulses, $t_1 < \tau/5$ (approximate as an impulse)
 - Long duration pulses, $t_1 > \tau/5$ (shape is considered)
- The main concern in shock is the maximum peak response-a measure of the severity under a shock.
- The shock response spectrum (SRS) is commonly used in industry.
- SRS is a plot of the maximum peak response of the single DOF spring-mass system.
- The peak is often labeled maxima.

Response of (short) rectangular excitation

$$u(t) = \frac{F_0}{K} (-\cos \omega_n t + \cos \omega_n (t-t_1)) \rightarrow \frac{u(t)K}{F_0} = -\cos \omega_n t + \cos \omega_n (t-t_1)$$

The response is plotted dimensionless, i.e.

$$xk/F_0 = (1 - \cos \omega_n t) - (1 - \cos \omega_n (t-t_1)) = -\cos \omega_n t + \cos \omega_n (t-t_1)$$

$$\left(\frac{xk}{F_0} \right)_{\max} = 2 \sin \frac{\pi t_1}{\tau}$$

$$= 2 \sin \frac{\pi}{8}$$

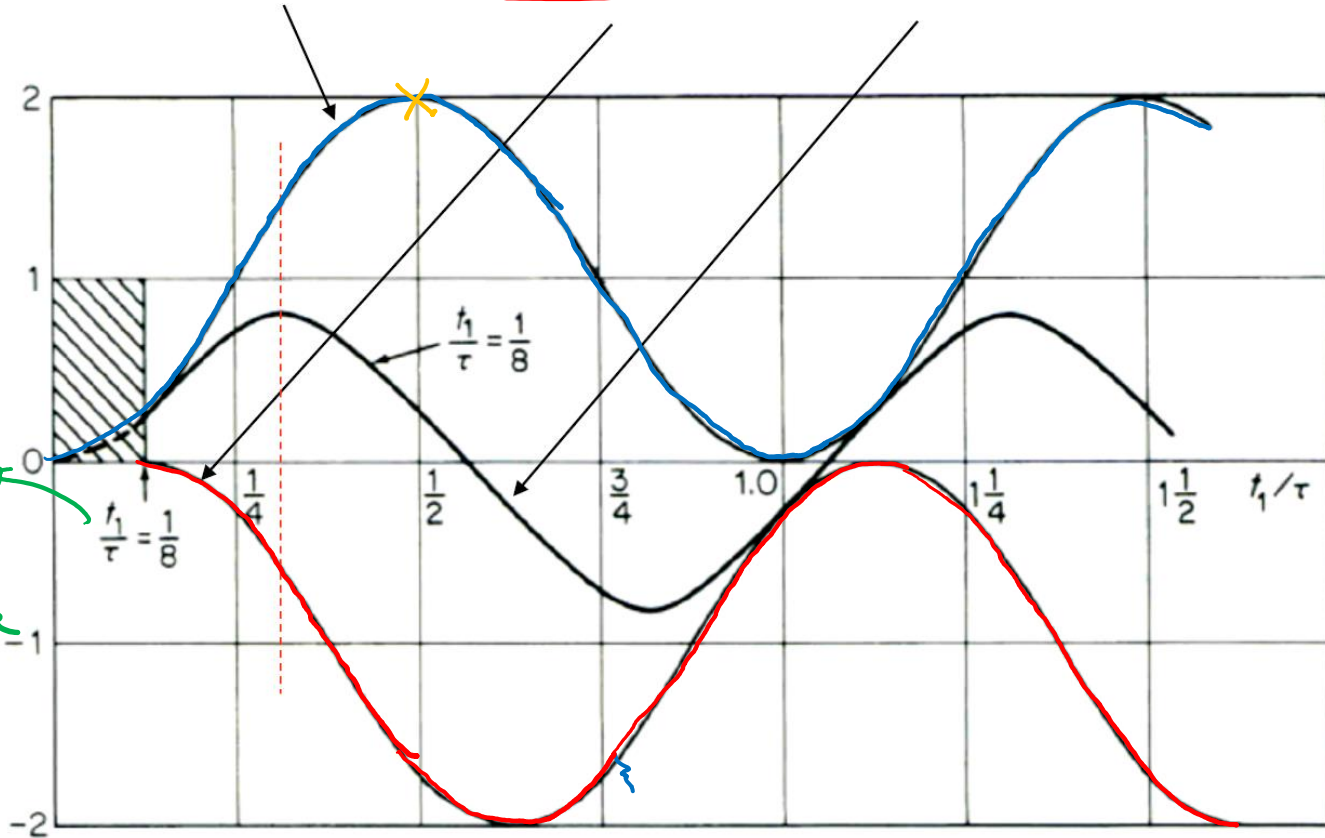
$$= 0.765$$

↑
Exact number

or

$$\left(\frac{xk}{F_0} \right)_{\max} < 1.25$$

$\frac{t_1}{\tau} = \frac{1}{8}$
 $\rightarrow t_1 = \frac{\tau}{8} < \frac{\tau}{5}$
 $\rightarrow$ short impulse



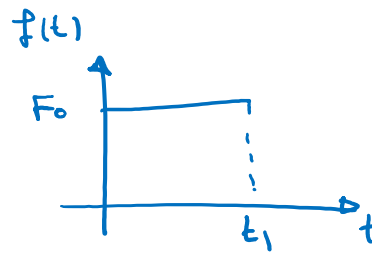
$$u(t) = 1 - \cos \omega_n t$$

$$\frac{t}{\tau} = 0.5 \rightarrow t = 0.5\tau$$

$$\rightarrow u(0.5\tau) = 1 - \cos \frac{2\pi}{\tau} \times 0.5\tau$$

$$1 - \cos \pi = 1 - (-1) = 2$$

Approximation



$$\text{if } t_1 < \frac{\tau}{5}$$

Recall the impulse response is

$$x = (1/m\omega_n) \sin\omega_n t$$

$$\text{Since } t_1 < \tau/5 = (2\pi/\omega_n)/5 = 1.25/\omega_n \rightarrow \omega_n < 2\pi/5t_1$$

and

$$I = F_0 t_1 < 1.25 F_0 / \omega_n$$

Then

$$x < 1.25 F_0 / m \omega_n^2 \sin\omega_n t = 1.25 F_0 / k \sin\omega_n t$$

$$\sin\omega_n t = 1 \rightarrow t_p = \pi/2\omega_n > (5/4)t_1 = (1/4)\tau$$

$$(xk/F_0)_{\max} < 1.25$$

↑ approximation for $u_{\max}$

$$t_1 < \frac{\tau}{5}$$

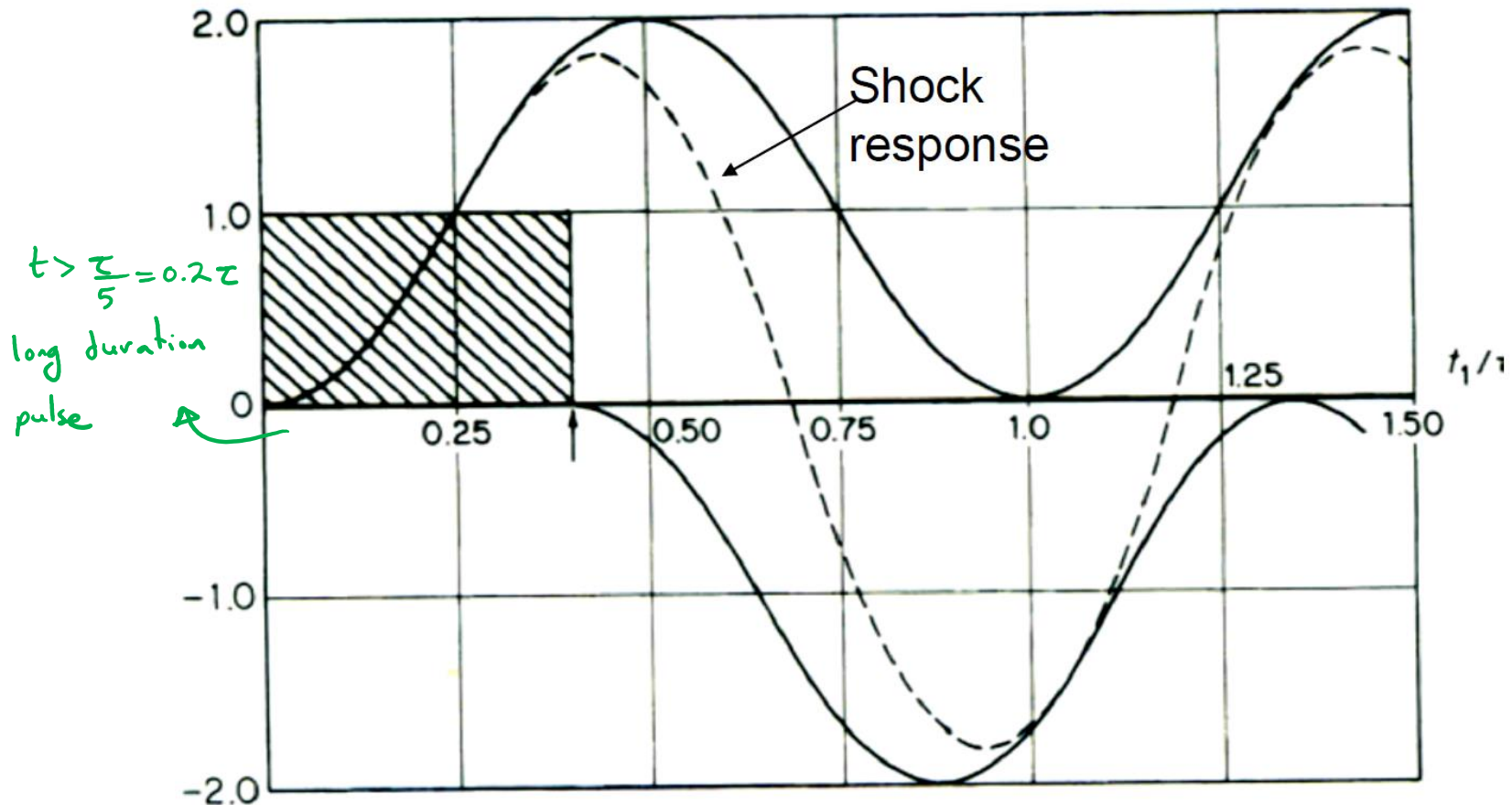
$$\omega_n = \frac{2\pi}{\tau} \rightarrow \tau = \frac{2\pi}{\omega_n} *$$

$$t_1 < \frac{\tau}{5}$$

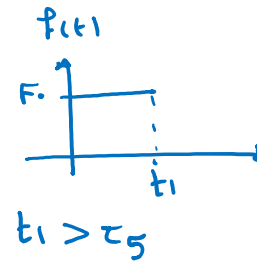
$$t_1 < \frac{2\pi}{5\omega_n}$$

$$\omega_n < \frac{2\pi}{5t_1}$$

Response of (long) rectangular excitation



Response peak



$$x(t) \Big|_{\max} ?$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

The response peak is determined by

$$d(xk/F_0)/dt = \omega_n(\sin \omega_n t_p - \sin \omega_n(t_p - t_1)) = 0$$

where t_p - time at the peak response.

Solve

$$\tan \omega_n t_p = -\sin \omega_n t_1 / (1 - \cos \omega_n t_1)$$

For which

$$\sin \omega_n t_p = \sin \omega_n t_1 / [2(1 - \cos \omega_n t_1)]^{1/2}$$

$$\cos \omega_n t_p = -[(1 - \cos \omega_n t_1)/2]^{1/2}$$

Then

$$(xk/F_0)_{\max} = [2(1 - \cos \omega_n t_1)]^{1/2} = 2 \sin(\omega_n t_1/2) = 2 \sin(\pi t_1/\tau) \quad (t > t_1)$$

$$\frac{x(t)k}{F_0} = -\cos \omega_n t + \cos \omega_n(t - t_1)$$

$$\left(\frac{xk}{F_0} \right)_{\max} = 2 \sin \left(\frac{\pi t_1}{\tau} \right)$$

Worst case scenario

$$\frac{t_1}{\tau} = 0.5 \rightarrow \tau = 2t_1$$

Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

$$F = kx \rightarrow \frac{xk}{F} = 1$$

Shock isolation

$$\left(\frac{xk}{F_0}\right)_{\max} = 2 \sin \frac{\pi t_1}{\tau}$$

- Shock isolation is to ensure that the maximum peak (transmissibility) must be less than unity.

$$2 \sin(\pi t_1/\tau) < 1 \quad \text{from } (xk/F_0)_{\max}$$

$$\rightarrow \sin(\pi t_1/\tau) < 0.5$$

$$\rightarrow \pi t_1/\tau < \pi/6 \quad (30^\circ)$$

Then

best case scenario $\tau > 6t_1$

$$t_1/\tau < 1/6 \rightarrow \textcircled{6t_1 < \tau} \quad \text{or} \quad \omega_n < \pi/3t_1 \rightarrow f_n < 1/6t_1$$

- The natural period of the isolated system must be six times greater than the pulse duration $\rightarrow$ the natural frequency.
- Other impulse shapes are bounded by a rectangle, and hence the rectangular excitation represents the upper bound.

$$\omega_n = \sqrt{\frac{k}{m}} \rightarrow \frac{2\pi}{\tau} = \sqrt{\frac{k}{m}} \xrightarrow{\tau > 6t_1} \frac{2\pi}{6t_1} = \sqrt{\frac{k}{m}} \rightarrow k ?$$

$$\rightarrow k = m \left(\frac{2\pi}{6t_1}\right)^2$$

Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

Damping effect $\rightarrow$ for an impulse

- Recall damped free vibration is

$$x = e^{-\xi\omega_n t} (C_1 \sin \omega_d t + C_2 \cos \omega_d t)$$

where $\omega_d = (1 - \xi^2)^{1/2} \omega_n$

$$C_1 = (v(0) + \xi\omega_n x(0)) / \omega_d ; C_2 = x(0)$$

Hence, the impulse response is

$$x = (v(0) / \omega_d) e^{-\xi\omega_n t} \sin \omega_d t$$

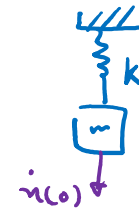
- t_p is determined by

$$dx/dt = 0 \rightarrow \tan(\omega_d t_p) = \omega_d / \xi\omega_n \text{ (assignment 4.1)}$$

$$t_p = (1/\omega_d) \tan^{-1}[(1 - \xi^2)^{1/2} / \xi]$$

- $X_{\max}$ is determined as

$$X_{\max} = (v(0) / \omega_d) e^{-\xi\omega_n t_p} \sin \omega_d t_p = 1 / m\omega_d e^{-\xi\omega_n t_p} \sin \omega_d t_p$$



Week # 8

$$x(t) = \frac{1}{m\omega_n} \sin \omega_n t$$

$$\text{or } x(t) = \frac{\dot{x}(0)}{\omega_n} \sin \omega_n t$$



Week # 8

$$* x(t) = \frac{\dot{x}(0)}{\omega_d} e^{-\xi\omega_n t} \sin \omega_d t$$

$$* \frac{dx(t)}{dt} = \frac{\dot{x}(0)}{\omega_d} \left[-\xi\omega_n e^{-\xi\omega_n t} \sin \omega_d t + \omega_d e^{-\xi\omega_n t} \cos \omega_d t \right]$$

$$\text{or } \tan \omega_d t_p = \frac{\omega_d}{\xi\omega_n}$$

Maximum displacement

- Since $t_p = (1/\omega_d) \tan^{-1}[(1-\xi^2)^{1/2}/\xi]$

$\rightarrow \tan(t_p \omega_d) = (1-\xi^2)^{1/2}/\xi$

- Let $\theta = \omega_d t_p = (1-\xi^2)^{1/2} \omega_n t_p$

$\tan \theta = (1-\xi^2)^{1/2}/\xi$

$\sin \theta = (1-\xi^2)^{1/2}$

- $X_{\max} = 1 / m \omega_d e^{-\xi \omega_n t_p} \sin \omega_d t_p = 1 / m \omega_d e^{-\xi \omega_n t_p} (1-\xi^2)^{1/2}$

$= 1 / m \omega_n e^{-\xi \omega_n t_p}$

$= 1 / m \omega_n e^{-\xi / (1-\xi^2)^{1/2} \text{Arctan}[(1-\xi^2)^{1/2}/\xi]}$

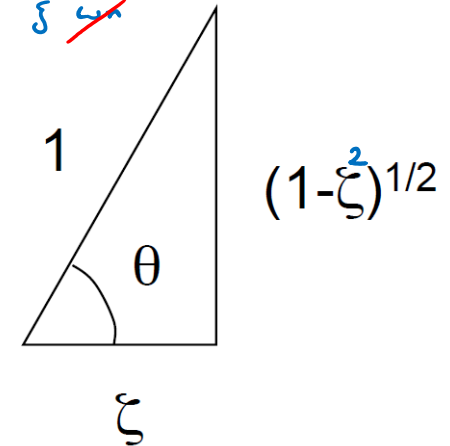
$= v / \omega_n e^{-\xi / (1-\xi^2)^{1/2} \text{Arctan}[(1-\xi^2)^{1/2}/\xi]}$

$\rightarrow X_{\max} = \frac{\dot{x}(0)}{\omega_n} e^{\quad}$

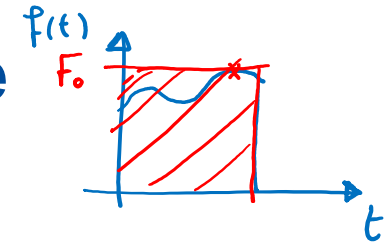
$$\tan \omega_d t_p = \frac{\omega_d}{\xi \omega_n}$$

$$\omega_d t_p = \text{Arctan} \frac{\omega_d}{\xi \omega_n}$$

$$t_p = \frac{1}{\omega_d} \text{Arctan} \frac{\sqrt{1-\xi^2} \omega_n}{\xi \omega_n}$$



Step response & rectangular pulse



- First, we look at the response of a spring-damper system to a step function. Then the response of a rectangular is considered as the sum of two step functions.

- The impulse response is

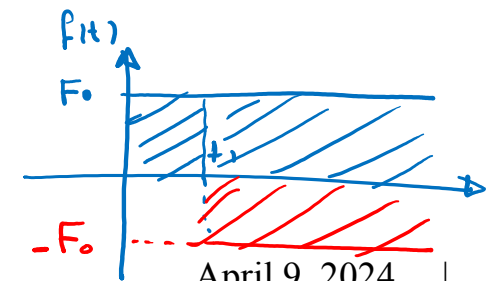
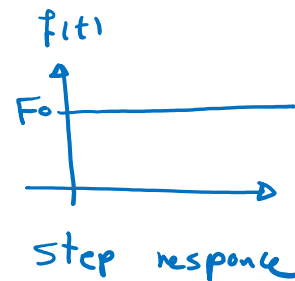
$$h = (v(0) / \omega_d) e^{-\xi \omega_n t} \sin \omega_d t = 1 / m \omega_d e^{-\xi \omega_n t} \sin \omega_d t$$

- Taking convolution leads to the step response

$$x(t) = \int_0^t F_o h(t - \xi) d\xi = (F_o / k) [1 - e^{-\xi \omega_n t} / (1 - \xi^2)^{1/2} \cos(\omega_d t - \psi)]$$

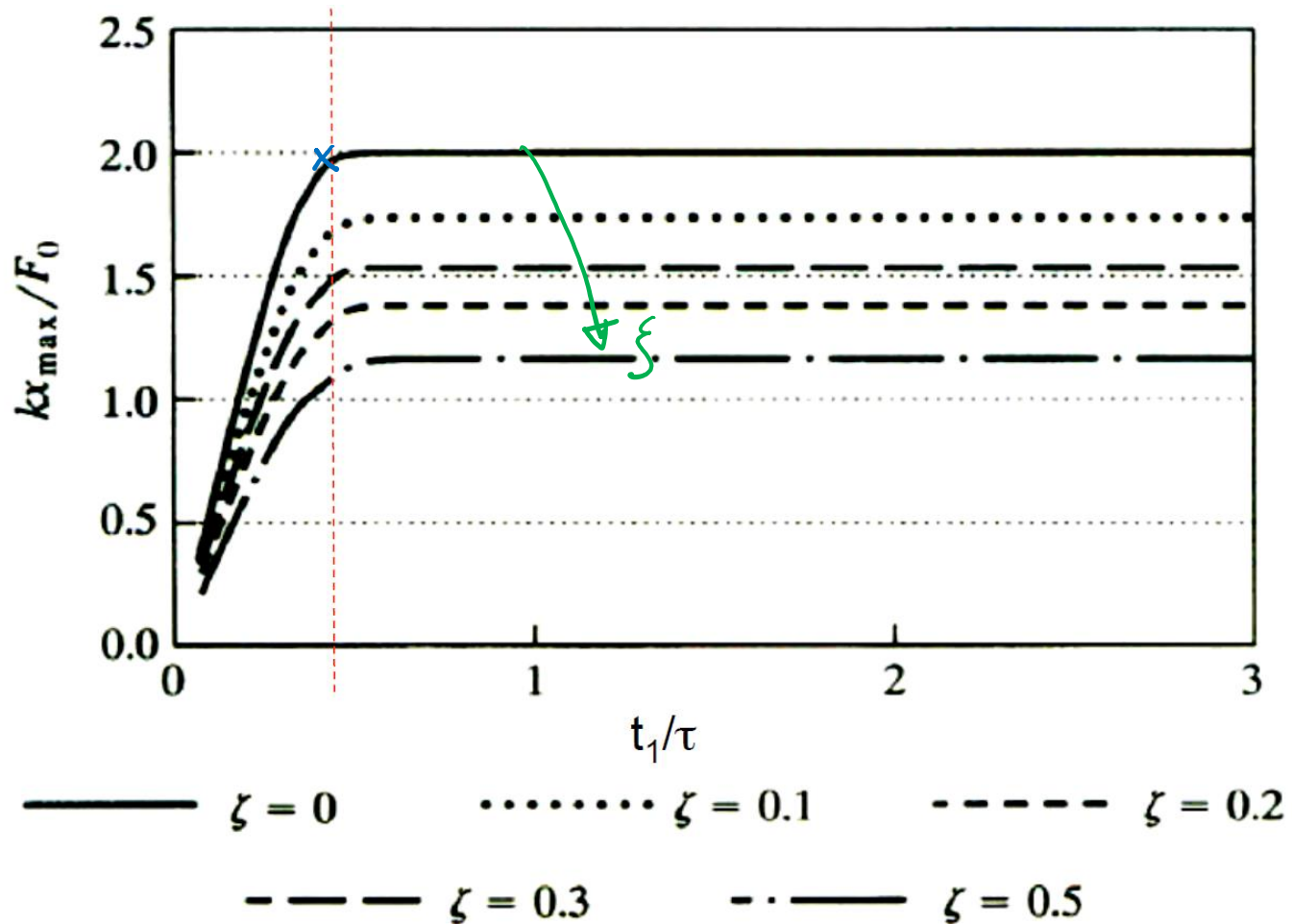
$$\tan \psi = \xi / (1 - \xi^2)^{1/2}$$

- It can be shown that $t_p = \pi / \omega_d$ (assignment 4.3)



the superposition of the two
will give the rectangular response.

Shock response spectrum (SRS)-Rectangular pulse



Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

Transmitting force

response to a step pulse

slide #26 $x(t) = \frac{F_0}{K} \left[1 - \frac{e^{-\xi \omega_n t}}{\sqrt{1-\xi^2}} \cos(\omega_d t - \psi) \right]$

The force transmitted through the spring-damper system is (short pulse)

$$F_T = kx + cdx/dt$$

→ $F_T = X' e^{-\xi \omega_n t} [k s \sin \omega_d t - c \xi \omega_n s \sin \omega_d t + c \omega_d \cos \omega_d t]$ ← transmitted force

Where $X' = (v(0)/\omega_d)$

Then $F_T = F' e^{-\xi \omega_n t} \sin(\omega_d t + \beta)$

Amplitude $F' = X' [(k - c \xi \omega_n)^2 + (c \omega_d)^2]^{1/2} =$

$$X' m \omega_n^2 = (v(0) / \omega_d) m \omega_n^2 = v(0) m \omega_n / (1 - \xi^2)^{1/2}$$

Phase $\beta = \text{Arctan}[(k - c \xi \omega_n) / c \omega_d]$

$$= \text{Arctan}[2 \xi (1 - \xi^2)^{1/2} / (1 - 2 \xi^2)]$$

Maximum transmitting force

The time at the maximum transmitting force is determined

$$t_{pf} = dF_T/dt = 0$$

→ $t_{pf} = (1/\omega_d) \text{Arctan}[(1-\xi^2)^{1/2}(1-4\xi^2)/\xi(3-4\xi^2)]$

time peak force

Maximum transmitting force is

$$F_{tmax} = F_T(t_{pf})$$

By examining $(1-\xi^2)^{1/2}$ and $(1-4\xi^2)$, for $0.5 \leq \xi < 1 \rightarrow t_{pf} = 0$

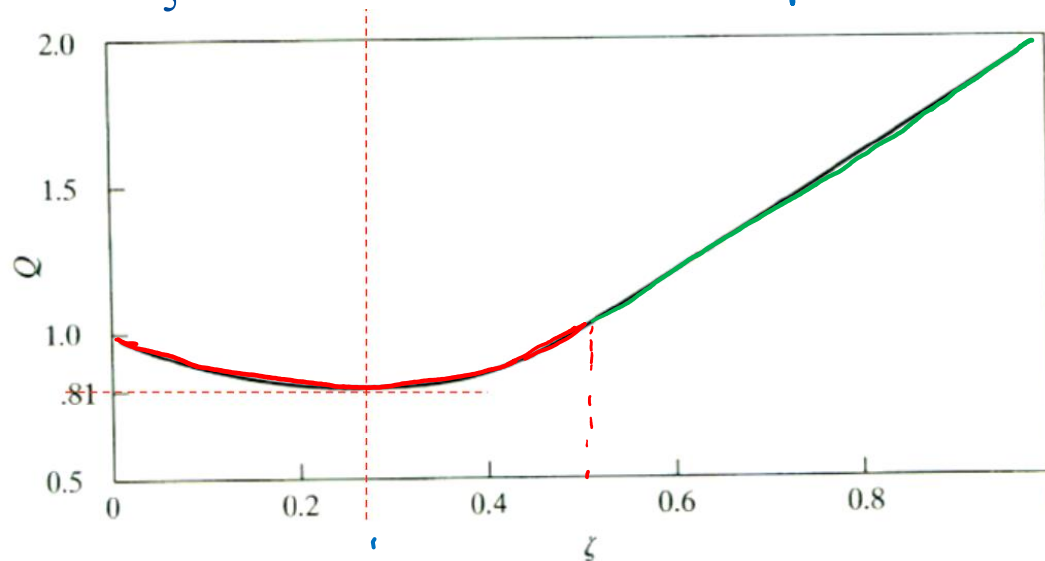
Force transmission ratio

- $Q(\xi)$ is defined as a non-dimensional measure of transmitting force

$$Q(\xi) = F_{\text{tmax}} / m\omega_n v$$

$$\begin{cases} e^{-\xi/(1-\xi^2)^{1/2}} \text{Arctan}[(1-\xi^2)^{1/2}(1-4\xi^2)/\xi(3-4\xi^2)] & \text{Branch 1} \\ 2\xi & \end{cases} \quad \begin{matrix} \xi < 0.5 \\ 0.5 \leq \xi < 1 \end{matrix}$$

- The smaller Q , the better, for shock isolation.
- Hence, optimal $\xi = 0.25 \rightarrow$ for transmitted force

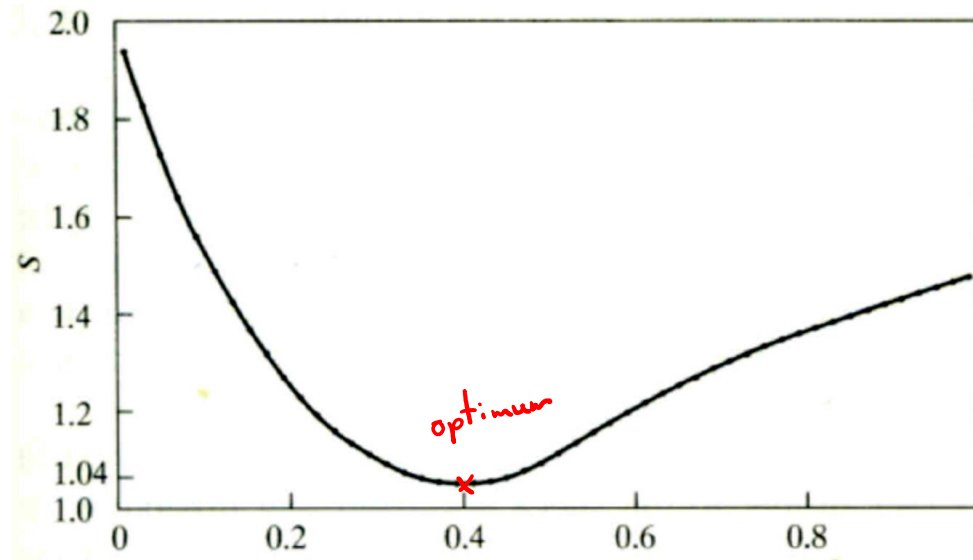


Energy absorption efficiency

- $S(\xi)$ is defined as a non-dimensional measure of energy absorption efficiency

$$S(\xi) = F_{T\max} x_{\max} / (mv^2/2) \rightarrow \text{absorber work/initial kinetic energy}$$

- The inverse of S represents the percentage of the absorbed energy v.s the work done by the absorber.
- The smaller S , the better, for energy absorption. Optimal $\xi=0.4$



for absorbed energy

Course outline

- Introduction
- Shock response spectrum
- Shock isolation
- Damping effect
- Transmitting force
- Energy method

Vibration assignment #4

Text: Chapter 4

Energy method

The energy generated during landing is

$$E = mV^2/2 + (mg - F_L)(S + S_t)$$

Where

m- aircraft weight (kg or lb) (Boeing 747-400 870K lbs)

V- sink speed (landing speed), 10ft/s, design weight, 6 ft/s max weight

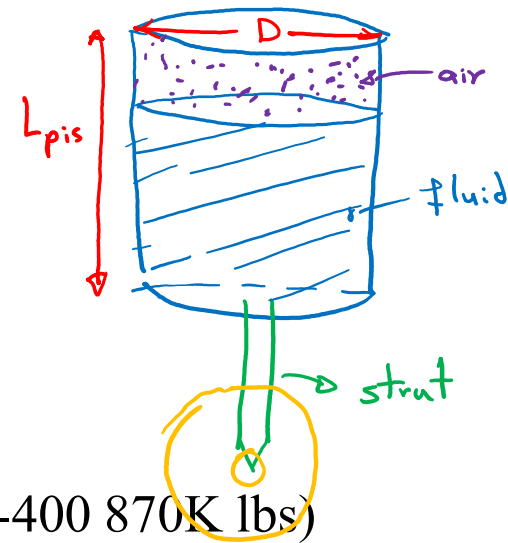
F_L - wing lift force (N, lbd)

S_t - tire deflection (mm, inch)

S- shock absorber stroke (mm, inch)

The work done by the shock absorber is expressed
$$W_s = \int_0^S F_T dx$$

Design is to satisfy $E = W_s$, so to determine stroke S



Industry design method

The design method used in industry is

$$(\eta_s S + \eta_t S_t) N m = m V^2 / 2 + (m g - F_I)(S + S_t)$$

Where

N- design factor (0.7-1.5), 1.2 is commonly used;

η_s - shock absorber efficiency factor, 0.47

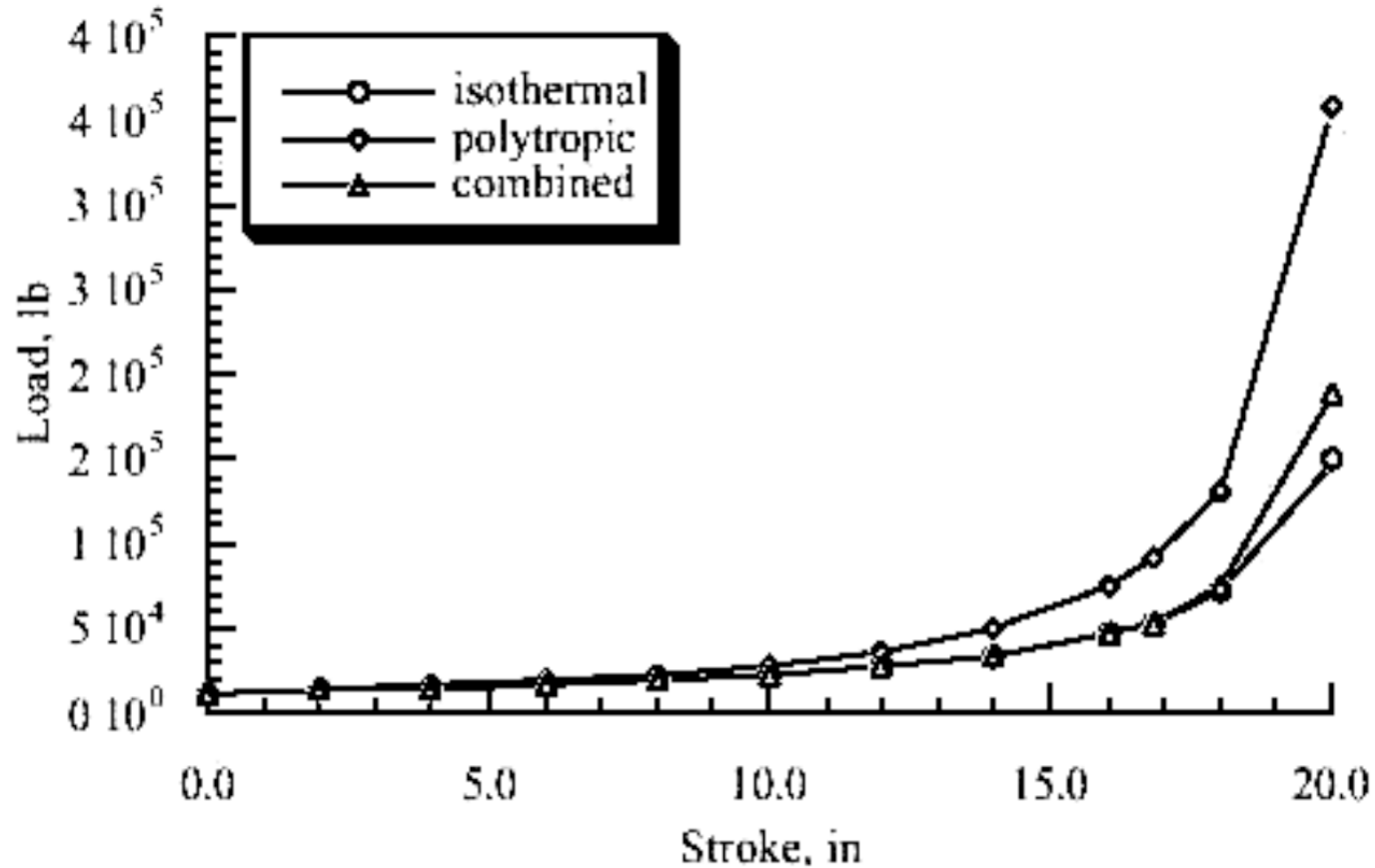
η_t - tire absorber efficiency factor, 0.8

In addition, the piston length is


$$L_{pis} = S + 2.75D$$

Where D- piston length

Stroke and load curve



Vibration assignment #4

Problems 4.1 – 4.4